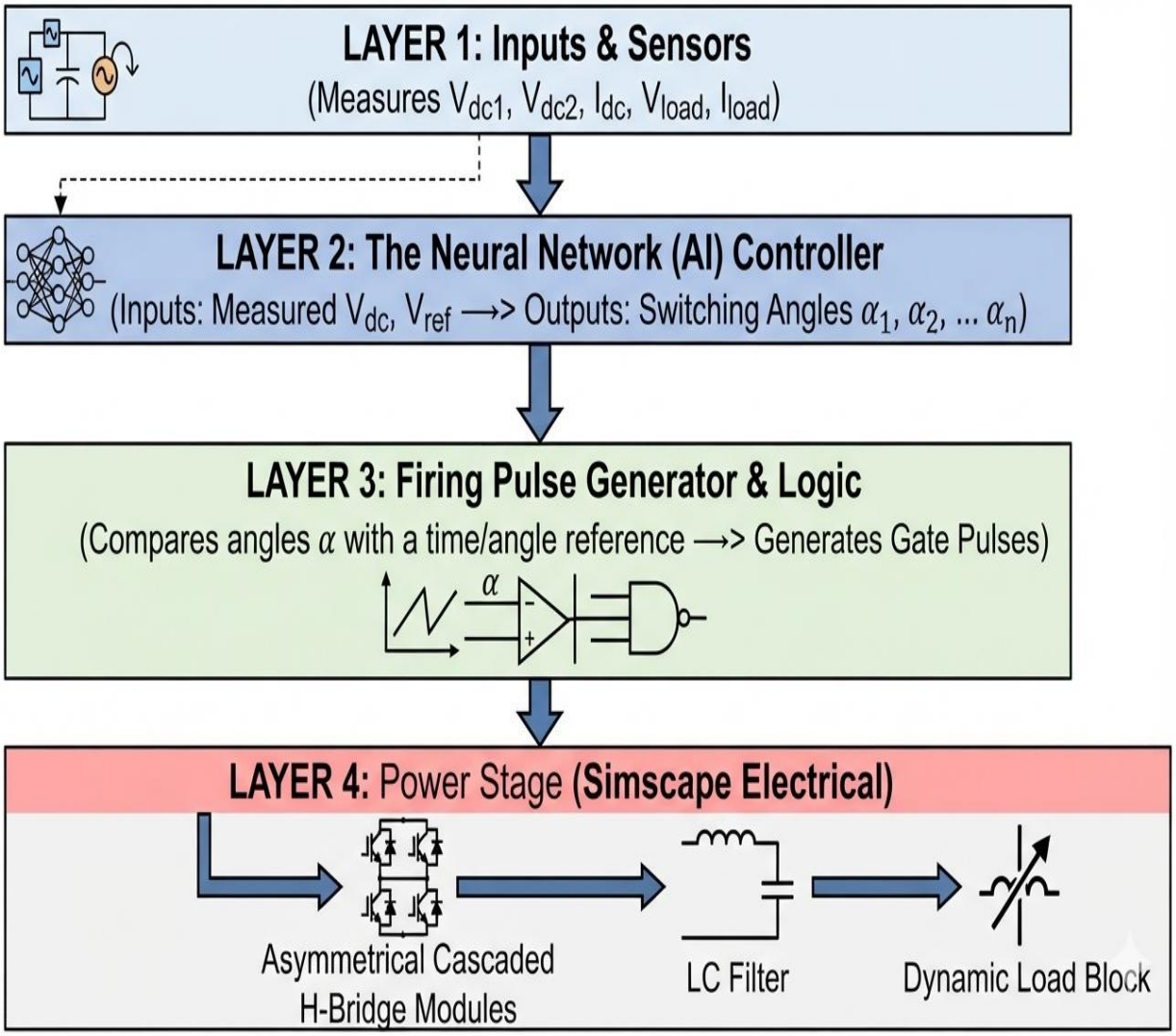
Machine Learning-Based Selective Harmonic Elimination for a Cascaded Multilevel Inverter under Variable DC-Link Conditions

Cascaded Multilevel Inverters (CMLIs) have been increasingly adopted in contemporary power supply systems because of their capacity to deliver high-quality power and minimize electromagnetic interference and operate at medium voltage levels. Nevertheless, low order harmonics suppression without causing high switching losses is a major challenge in these systems. One of the most efficient modulation strategies in addressing this problem is Selective Harmonic Elimination (SHE), which requires solving complex non-linear transcendental equations. Classic approaches use offline calculations with lookup tables that do not allow to take into account the dynamics of the system. For instance, the changing DC-link voltage, which is typical for solar PV generators, and rapidly changing dynamic loads cannot be incorporated into these classic approaches.

This work proposes an approach to designing a real-time ANN-based SHE controller in MATLAB/Simulink. Firstly, a mathematical model of a highly asymmetrical cascaded inverter is developed, and a huge amount of optimal switching angles for different values of modulation index is computed numerically. Then, the Deep Learning Toolbox of MATLAB is used to train an ANN that can find the required switching angles instantaneously from input parameters.

The whole system is simulated in Simulink using Simscape Electrical. A trained machine learning block is included into the gate-control loop of the inverter. To prove the efficiency of the suggested approach, several transient simulations with severe test scenarios, including DC-link voltage degradation (partial solar shading) and non-linear load stepping, are performed. It is expected that the ANN-based SHE controller will reach real-time convergence in less than 1 ms, keep the phase voltages balanced and reduce Total Harmonic Distortion (THD) well below the 5% IEEE 519 standard threshold.

Simulink Block Diagram Architecture



1. Simulink Block Diagram Architecture

The proposed framework maps a digital controller layer processing machine learning inference directly to an electrical network modeled within Simscape Electrical. The system operates on a four-tier architecture sequence:

Layer Block	Target Parameters & Functionality
Layer 1: Inputs & Sensors	Measures real-time system behaviors including input source variables (V_{dc1}, V_{dc2}, I_{dc}) and load state metrics (V_{load}, I_{load}).
Layer 2: AI Controller	Accepts measured DC link levels and the tracking target framework (V_{ref}). Maps variables instantly to compute optimized switching angles (α_1, α_2, ... α_n) via an integrated artificial neural network template.
Layer 3: Pulse Generator	Maintains a phase accumulator reference synchronized to the grid loop. Uses comparison operations to produce clean gate firing signals based on the dynamic α parameters.
Layer 4: Power Stage	Modeled inside Simscape Electrical. Incorporates asymmetric Cascaded H-Bridge cells, secondary structural LC smoothing filters, and a dynamic three-phase step load block.

2. Step-by-Step Simulink Implementation Plan

Phase 1: Mathematical Setup & Data Generation

- Formulate standard non-linear transcendental Fourier expressions representing the primary operational voltage curve and desired harmonic cancellation targets (5th, 7th, 11th).
- Build a numerical execution template in MATLAB utilizing optimization routines (e.g., *fsolve* or global solvers).
- Sweep modulation indexes spanning 0.1 to 1.0 along with cross-variable asymmetric DC unbalances to establish a reference data array.

Phase 2: AI Training & Block Deployment

- Load optimization parameters inside the Deep Learning Toolbox setup.

• Construct a feedforward tracking layout using two hidden networks containing hyperbolic tangent sigmoid activation functions.

• Train utilizing the Levenberg-Marquardt algorithm until performance reaches desired Mean Squared Error limits ($< 10^{-5}$).

• Generate native Simulink execution blocks seamlessly using the *gensim(net)* function tool.

Phase 3: Simscape Circuit Assembly

• Assemble structural H-Bridge array paths inside the Simscape block space utilizing individual power semiconductor switches.

• Link dynamic signal-controlled sources to represent physical PV input degradation conditions.

• Structure three-phase dynamic breakers capable of applying load step disturbances during active execution timelines.

Phase 4: Integration & Performance Check

• Route the compiled AI computation block into the primary gate driver comparison loops.

• Adjust simulation step properties to fixed-step mode using an analytical solver tailored for stiff systems (e.g., *ode14x* or discrete solvers) at a resolution of 1 microsecond.

• Extract active system vectors post-run and execute fine harmonic checks using the Powergui FFT Analysis Tool to evaluate standard compliance (IEEE 519 standard).